GAME DESIGN & LEVEL SPECIFICATION DOCUMENT (GDD)

# Still Life

First-Person Environmental Puzzle & Companion Gating Design Specification

| | | | |
|---|---|---|---|
| **Author / Role** | Kevin (Lefi) Shan: Solo Game & Level Designer, Programmer | **Engine & Stack** | Unity 6 (URP), C#, Custom HLSL Kuwahara/Ink Shaders |
| **Production Scope** | Solo Prototype · USC CTIN (Situated Role: The Dreamer) | **Facility Scope** | 200m Linear Facility (5 Rooms + 3 Hallways) · 4-Phase Weight Locks |



**Figure 1.1:** Room 2 Three-Story Archive where animated book flocks flutter past animal vaults.

### Executive Summary & Level Design Intent

Still Life is a first-person puzzle game set inside a monochrome gallery and foundry where the behaviors of animals and household objects are swapped. Animals stand still and silent on pedestals, so players use them as pressure-plate counterweights, bridges, and gear wedges. Meanwhile, a desk lamp shivers in the dark, flees when startled, and warms up when petted.

Across Rooms 1 to 3, every lock-and-key puzzle trains the player to treat animals strictly as physical tools while escorting the lamp. In Room 4, an industrial shredder blocks the exit and prompts for one object. When the shredder rejects the dog as a living organism, the player's own habit triggers the twist without a single line of dialogue.

## 1. Core Level Design Pillars & Puzzle Framework

Section 1.1 to 1.3

### 1.1 Multi-Use Animal Props (5 to 100 kg)

To keep pressure-plate math readable without messy physics stacking, every animal is tuned to a multiple of 5 kg and serves three mechanical roles: counterweights on brass plates, bridges across 4.2m floor gaps, and wedges jammed into machinery.

### 1.2 Dual-Agent Soft & Hard Gating

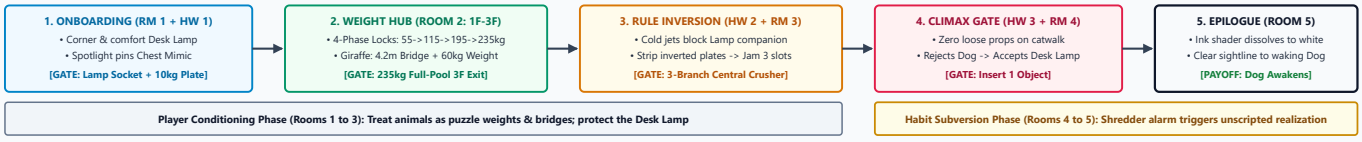
Every room exit is a hard gate requiring both the player and the desk lamp. Territorial chest mimics and cold air vents act as companion-only soft gates: the player can walk past them, but the lamp stops until the player routes weights to clear the hazard.

### 1.3 Zero-UI Spatial Readability

Right-angle floor and wall indicator lines connect every plate to its target device by color (Cyan for gates/lifts, Gold for spotlights, Green for crusher jammers), keeping three-story puzzle dependencies readable without HUD markers.

**Figure 1.2: Facility Pacing, Gating & Mechanic Progression Graph (Rooms 1 to 5)**

Corridors Alternating with Multi-Story Hubs




## 2. Animal & Companion Weight Specification Catalog

Table 2.1: 5 kg Increments vs. 1.4 kg Blueprint Clue

| Catalog ID | Prop / Entity | Weight | Level Design Role | Puzzle Function Across Rooms 1 to 5 |
|---|---|---|---|---|
| `SPEC-01` | **Chicken** | `5 kg` | Fine Weight Unit | Pairs with Cat (10 kg) to reach 15 kg plate thresholds in Room 2 and Room 3. |
| `SPEC-02` | **Cat** | `10 kg` | Light Counterweight | Powers the Hallway 1 Spotlight Stand (10 kg) and combines with Chicken for 15 kg locks. |
| `SPEC-03` | **Penguin** | `15 kg` | Single-Prop 15 kg Key | Unlocks the 2F Dog Vault (15 kg), 1F SE Elevator (15 kg), and 15 kg Spotlight Plates. |
| `SPEC-04` | **Dog** | `25 kg` | Mid Weight & Finale Subject | Opens 25 kg gates in Rooms 2 and 3; rejected by Room 4 shredder before waking in Room 5. |
| `SPEC-05` | **Deer** | `40 kg` | Phase 3 Vault Key | Unlocked in Room 2 NW Vault; holds the 40 kg 3F Spotlight Plate for the exit sequence. |
| `FIG. 02` | **Giraffe (Blueprint #2)** | `60 kg` | 4.2m Bridge & Counterweight | Tips across 4.2m floor gaps on 2F and 3F as a bridge, or weighs 60 kg on plates. |
| `FIG. 03` | **Tiger (Blueprint #3)** | `80 kg` | Lift Weight & Gear Wedge | Powers the 80 kg 2F-to-3F Elevator in Room 2 and jams the mid-tier crusher gears in Room 3. |
| `FIG. 01` | **Horse (Blueprint #1)** | `100 kg` | Corridor Blocker & Wedge | Rises in Room 1 to corner the fleeing lamp, and jams the upper flywheel in Room 3. |
| `FIG. 04` | **Desk Lamp (Blueprint #4)** | `1.4 kg` | **Companion & Climax Key** | **Lights dark areas and powers exit sockets; blueprint note clues that sensors read it as an object.** |

## 3. Master Facility Layout (Z = -12m to +188m) & Room 1 Onboarding Flow

5 Exhibition Rooms & 3 Threshold Hallways

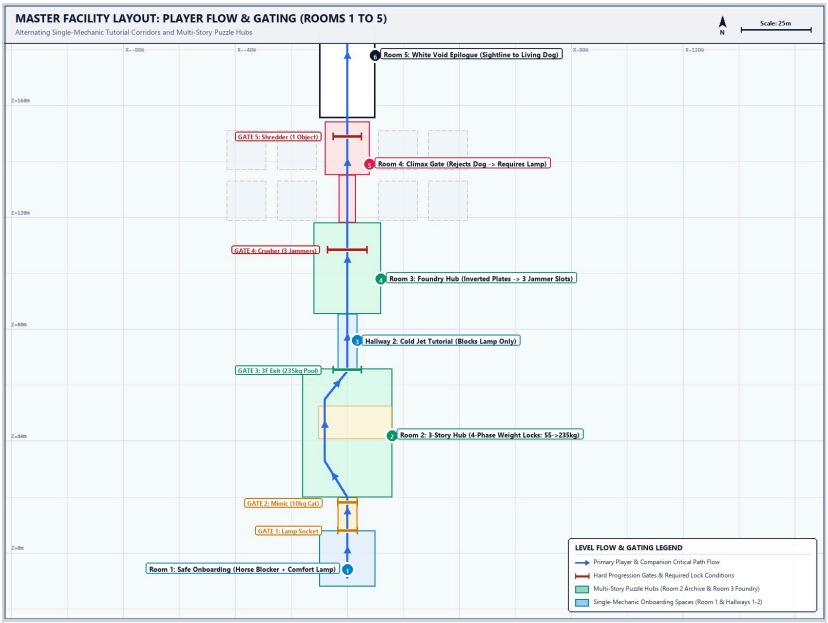



**Figure 3.1: Master Facility Flow & Gating Map (Rooms 1 to 5):** Orthographic layout showing the primary player and companion path (blue arrows) and the 5 hard progression gates.

### 3.1 Pacing via Corridors and Multi-Story Hubs

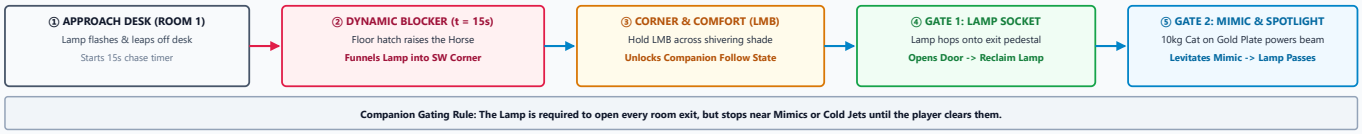
The 200m linear spine alternates between compact corridors that introduce one mechanic in isolation and multi-story hubs that combine those mechanics vertically:

- **Room 1 & Hallway 1 (Z = -12m to +20m):** Introduces lamp comforting, optical exit sockets, and spotlight beams that pin chest mimics.
- **Room 2 Archive Hub (Z = +20m to +66m):** Three-story puzzle hub (1F, 2F, 3F) with 7 animal vaults and a 4-phase weight lock sequence.
- **Hallway 2 & Room 3 Foundry (Z = +66m to +118m):** Introduces cold air jets in Hallway 2, then inverts plate rules in Room 3 to jam the central crusher.
- **Hallway 3, Room 4 & Room 5 (Z = +118m to +188m):** Narrows into an isolated catwalk before the shredder gate and pure white epilogue.

### 3.2 Dynamic Choke Point in Room 1 Onboarding

When the player approaches the desk in Room 1, the startled lamp flashes brightly and hops onto the floor. In early playtests, players chased the lamp in circles around the room. To solve this spatially without UI prompts, a floor hatch opens after 15 seconds and raises the Horse across the main lane, funneling the lamp into the SW corner where the player can corner and comfort it.

**Figure 3.2: Room 1 & Hallway 1 Spatial Flow, Dynamic Choke Point & Onboarding Gates** Corner Choke Point -> Lamp Socket -> Spotlight Gate




## 3.3 Visual Indicator Lines, Shader Contrast & Static Mesh Optimization

Table 3.1: Readability & Rendering Systems

| System / Shader Module | Target Level Element | Level Design Readability & Technical Purpose |
|---|---|---|
| `ConduitDirectionalPulse.shader` | Floor & Wall Indicator Lines | Connects every pressure plate to its target device using 3 colors: Cyan (doors/lifts), Gold (spotlight stands), and Green (crusher jammers). |
| `AnimalInkShader.shader` | Animal Counterweights | Triplanar ink cross-hatching and flat matte shading with zero idle animation, reinforcing the player assumption that animals are static puzzle props. |
| `LivingApplianceCreature.shader` | Desk Lamp & Chest Mimics | Hand-drawn outline boil, shivering displacement in dark or cold zones, and warm subsurface glow when comforted by the player. |
| `Room4AvantGardeInfiniteBuilder` | Room 4 Repeating Grid | Merges hundreds of surrounding repeating gallery cells into 6 static UInt32 meshes (~6 draw calls) to maintain 200+ FPS. |



**Figure 3.3: Room 1 Dynamic Blocker:** The Horse rises from the center floor hatch to block the loop and corner the fleeing lamp.

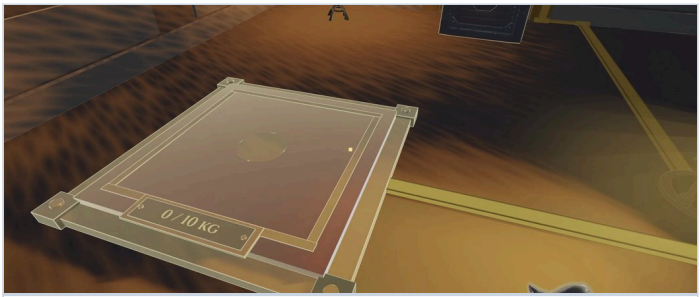



**Figure 3.4: Hallway 1 Spotlight Gate:** Placing the 10 kg Cat on the gold plate powers the spotlight beam, levitating the chest mimic.

## 4.1 Room 2 (Three-Story Archive): 4-Phase Weight-Lock Progression

`1F / 2F / 3F (0 kg Leftover)`

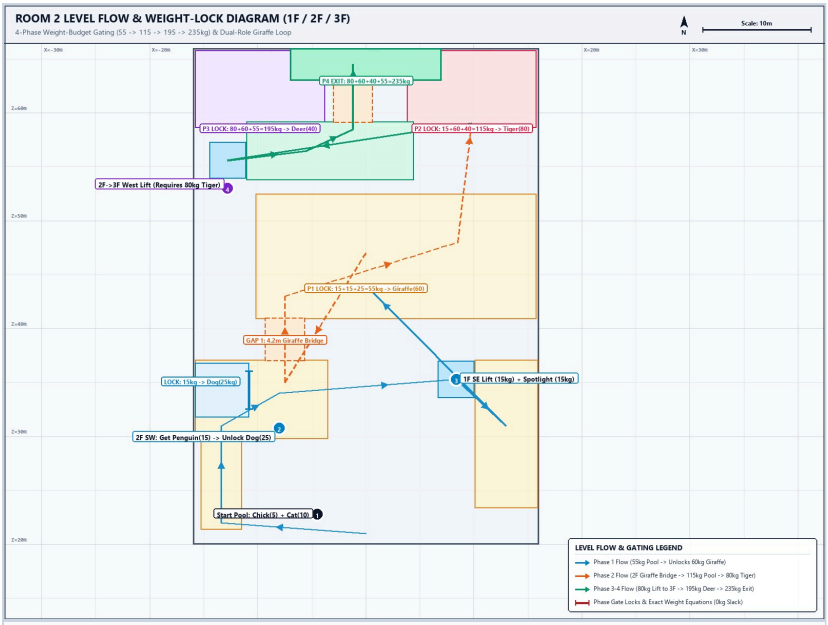



**Figure 4.1: Room 2 Flow & Weight-Lock Floorplan (1F / 2F / 3F):** Showing Phase 1 to 4 traversal loops, the 2F and 3F 4.2m Giraffe bridges, and exact weight-lock equations.

### Preventing Sequence Breaks: 100% Weight-Pool Allocation

In multi-prop weight puzzles, unlocking heavier weights often leaves spare mass that lets players skip earlier locks. Room 2 prevents this by requiring 100% of the currently unlocked weight pool at the peak of every phase (55 kg -> 115 kg -> 195 kg -> 235 kg). With 0 kg left over, players must constantly reclaim and re-route every animal across all three floors.

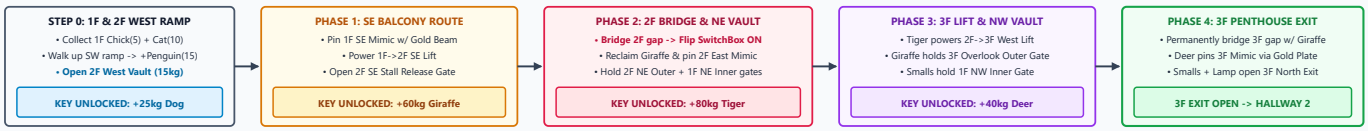
### Multi-Use Prop: The 60 kg Giraffe Bridge & Counterweight

Unlocked at the end of Phase 1, the 60 kg Giraffe serves two spatial roles: its long neck tips across the 4.2m floor gaps on 2F and 3F as a bridge, or reclaims into inventory after flipping the 2F bypass switch to act as a 60 kg counterweight on vault plates.

### Preventing 3F Soft-Locks & Sequence Skips

Without unlocking the 40 kg Deer in Phase 3, the player has only 195 kg against the 235 kg 3F exit requirement. Once the Deer is unlocked, tipping the Giraffe across the 3F gap locks it permanently in place so players cannot pick it back up on the far side and strand themselves on the top floor.

**Figure 4.2: Room 2 Multi-Floor Lock-and-Key Dependency Flow (1F <-> 2F <-> 3F)** How Each Unlocked Animal Opens the Next Vertical Route




## 4.1.2 Four-Phase Weight Allocation Table & In-Game Execution

`Table 4.1 & Figures 4.3 to 4.5`

| Phase | Unlocked Pool | Simultaneous Active Locks (100% Pool Utilization) | Leftover & Reward |
|---|---|---|---|
| **Phase 1** | `55 kg` (5+10+15+25) | `-15 kg` 1F Spotlight (5+10) + `-15 kg` SE Lift (15) + `-25 kg` 2F SE Gate (25) = **`55 / 55 kg`** | `0 kg` -> **+60 kg Giraffe** |
| **Phase 2** | `115 kg` (55 + 60) | After 2F bridge switch: `-15 kg` 2F Beam + `-60 kg` NE Outer + `-40 kg` NE Inner = **`115 / 115 kg`** | `0 kg` -> **+80 kg Tiger** |
| **Phase 3** | `195 kg` (115 + 80) | `-80 kg` 2F->3F Lift (Tiger) + `-60 kg` 3F Outer Gate (Giraffe) + `-55 kg` 1F NW Inner Gate = **`195 / 195 kg`** | `0 kg` -> **+40 kg Deer** |
| **Phase 4** | **`235 kg` (All 7)** | `-80 kg` 2F->3F Lift + `-60 kg` 3F Gap Bridge + `-40 kg` 3F Spotlight + `-55 kg` 3F Exit Gate = **`235 / 235 kg`** | `0 kg` -> **Exit Open** |




**Figure 4.3: 3-Story Atrium:** Sightlines across 1F vaults and 2F/3F balconies.




**Figure 4.4: Carrying 60 kg Giraffe:** Routing the Giraffe past the 1F spotlight beam.

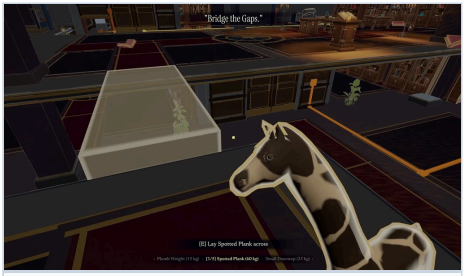



**Figure 4.5: 4.2m Giraffe Bridge:** Tipping the Giraffe across the 2F floor gap.

## 4.2 Hallway 2 & Room 3: Cold Air Gates, Inverted Plates & Crusher Lock

Rule Inversion & 3-Spoke Hub Jamming

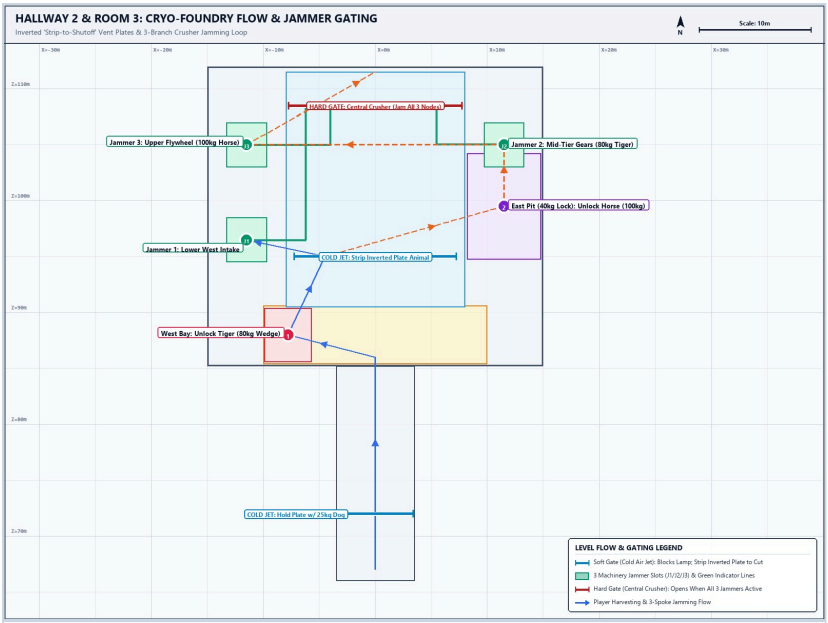



**Figure 4.6: Hallway 2 & Room 3 Flow & Jammer Gating Map:** Showing Cold Air Jet soft gates, Inverted Vent Plates, the 3 Jammer Slots (J1, J2, J3), and green indicator lines to the Central Crusher.

### Hallway 2 Onboarding: Companion-Only Cold Air Gates

Hallway 2 introduces cold air vents that blow freezing streams across the corridor. While the player can walk straight through the air stream unharmed, the desk lamp shivers and stops at the edge of the jet. To escort the lamp across, the player must hold down the shutoff plate with the 25 kg Dog.

### Subverting Plate Habits: Inverted "Strip-to-Shutoff" Plates

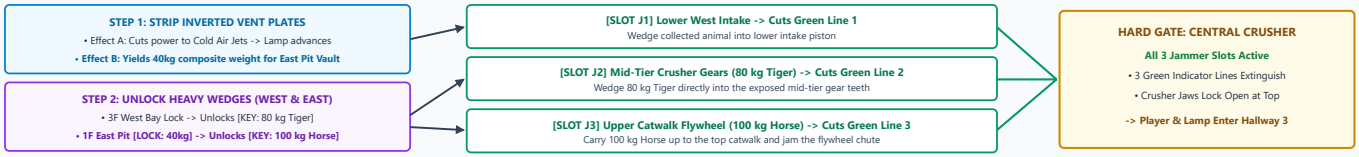
In Rooms 1 and 2, placing weight on a plate always opened a path. Room 3 flips this expectation: pre-placed animals sit on inverted power plates that keep the room's freezing jets running. Picking an animal up off its plate shuts off the cold air jet blocking the lamp and adds that animal to your inventory for the crusher puzzle.

### 3-Branch Central Crusher Gate & Jammer Slots

The exit to Room 4 is blocked by a central mechanical crusher fed by three green floor indicator lines. Rather than holding plates, players permanently jam three sub-mechanisms using harvested animals: (J1) Lower West Intake, (J2) Mid-Tier Crusher Gears (jammed with the 80 kg Tiger), and (J3) Upper Flywheel (jammed with the 100 kg Horse).

**Figure 4.7: Room 3 Lock-and-Key Dependency Chart (Inverted Plates -> 3 Jammer Slots -> Crusher Gate)**

Hazard Removal + Heavy Prop Harvesting




### 4.2.3 In-Game Execution: Cold Air Vents, Central Crusher, Tiger Gear Jam & Horse Flywheel Lock

Figures 4.8 to 4.11




**Figure 4.8: Hallway 2 Cold Vent:** Holding the shutoff plate so the lamp can cross.

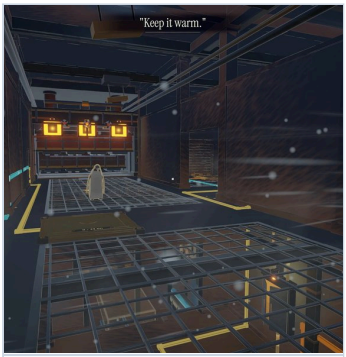



**Figure 4.9: Central Crusher Gate:** Fed by 3 green floor indicator lines.




**Figure 4.10: Jammer Slot 2 (Tiger):** Wedging the 80 kg Tiger into the gears.

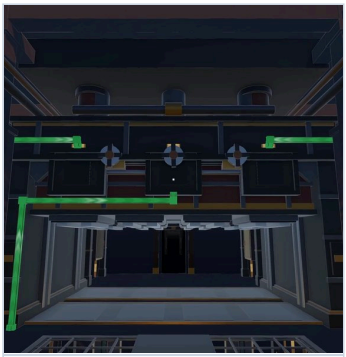


**Figure 4.11: Jammer Slot 3 (Horse):** Locking the upper flywheel cuts the last line.

## 4.3 Room 4 & 5: Shredder Gate Habit Subversion & Room 5 Sightline Payoff

`Sensor Rejection & Finale Sightline`

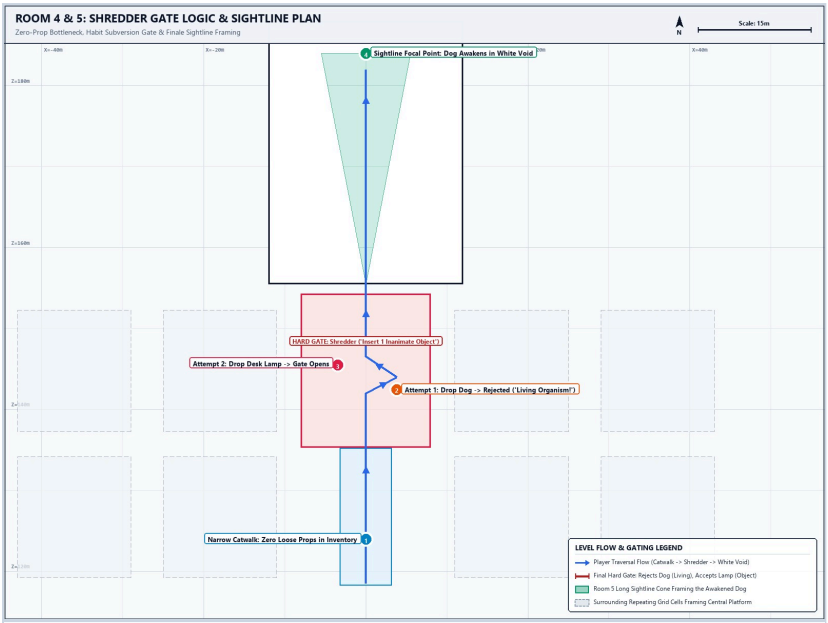



**Figure 4.12: Room 4 Shredder Gate & Room 5 Sightline Plan:** Showing the narrow catwalk, the shredder rejection/acceptance loop, and the clear sightline to the awakened dog.

### Subverting Player Conditioning at the Shredder Gate

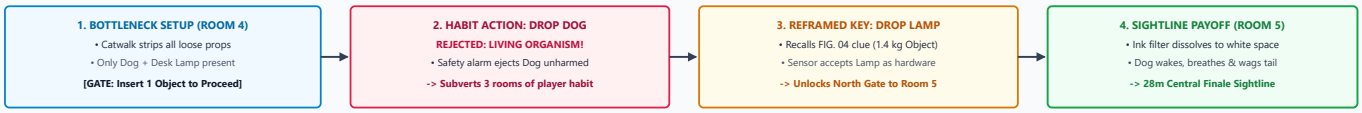
Before entering Room 4, Loading Screen #4 displays blueprint FIG. 04 (DESK LAMP, 1.4 KG): "It shivers in the dark and leans into your hand like something alive, but to the gallery's sensors it is still just an object." Inside Room 4, a narrow steel catwalk with zero loose props leads to an industrial shredder blocking the exit with the prompt: INSERT 1 OBJECT TO PROCEED. Beside the hopper sits the dog.

Conditioned by three rooms of jamming animals into machinery, players instinctively drop the dog into the chute. Instead of crushing it, the shredder blares "SAFETY MECHANISM ACTIVATED: LIVING ORGANISM DETECTED!" and ejects the dog unharmed. With no other loose props in the space, the player realizes that the silent animals were alive all along and the desk lamp is the only object the sensor will accept.

### Room 5: Visual Filter Dissolve & Long Sightline Payoff

Dropping the desk lamp into the shredder opens the north gate into Room 5. Stepping across the threshold dissolves the monochrome ink shader into a pure white space, framing a 28m central sightline to Z = +184m where the dog wakes up, breathes, perks its ears, and wags its tail.

**Figure 4.13: Room 4 Shredder Gate Decision Tree & Habit Subversion Flow** Dog Rejected -> Lamp Accepted -> Room 5 Payoff



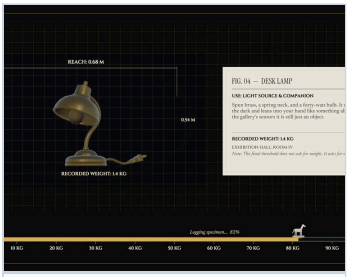




**Figure 4.14: FIG. 04 Blueprint Clue:** "To the sensors it is just an object."

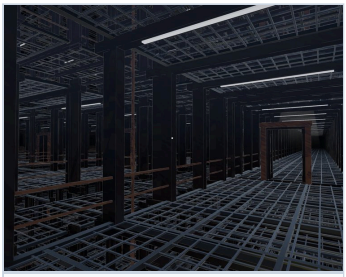


**Figure 4.15: Dog Rejected:** "LIVING ORGANISM DETECTED!"

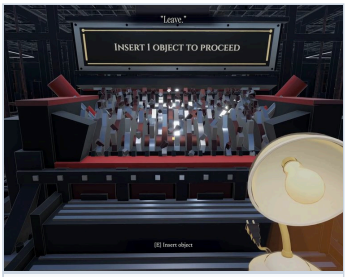



**Figure 4.16: Lamp Accepted:** Holding the lamp at the shredder chute.



**Figure 4.17: Room 5 Sightline:** The dog wakes and wags its tail.

## 5. Level Design Retrospective & Key Takeaways

`Post-Mortem Synthesis`

### 5.1 Conditioning Through Repetition

Rather than delivering its theme through exposition, the facility uses 30 minutes of weighing, bridging, and jamming animals to build an automatic reflex, so the player's own action triggers the Room 4 reversal.

### 5.2 Zero-Leftover Weight Math Prevents Skips

Requiring 100% of the unlocked weight pool at each phase of Room 2 (55 -> 115 -> 195 -> 235 kg) ensured that unlocking heavier animals never made lighter ones obsolete; every prop had to be reclaimed across all three floors.

### 5.3 Dynamic Geometry Over Tutorial Popups

Raising the Horse in Room 1 to corner the fleeing lamp and routing color-coded indicator lines across floors solved early playtest confusion through level geometry rather than UI text.